

Literature Review

Sl#	Article	Dataset	Primary Contribution	Limitations
1	A Digital Twin-Based Intelligent Network Architecture for UASNs	Simulated Aqua-Sim Twin (4th Gen) & tank data.	Proposes DTNA separating local node-level optimization (RAPD) from global synchronization (CMFD).	Tested primarily in simulation/tanks; lacks deep-sea validation.
2	DTEAR: Digital Twin-Based Energy-Efficient Adaptive Routing for UASNs	Adapted C-Town water distribution benchmark.	Integrates digital twins with depth-based routing to preserve link quality and minimize query overhead.	Assumes stable link availability to surface nodes; uses terrestrial benchmark.
3	Cognition Difference-Based Dynamic Trust Network for Distributed Bayesian Data Fusion	Simulated distributed sensor data (IROS).	Builds a dynamic trust network to improve how Bayesian data fusion weighs each sensor node's input.	Not tied to specific deep-water physics; generalized methodology.

4	Anomaly Detection of Underwater Gliders Verified by Deployment Data	Real-world glider deployment telemetry (USF Stella/SkIO Angus).	Verifies anomaly detection algorithms using actual ocean deployment logs.	Focuses largely on mechanical anomalies rather than acoustic wake events.
5	Deep-sea hydrophones: review of macro-structured and new MEMS hydrophones	Device specifications and literature review.	Comprehensive review of emerging MEMS hydrophone technology for deep-sea acoustic detection.	Purely a hardware review without an algorithmic software framework.
6	A Sensor-Based TinyML Acoustic Monitoring System...	Terrestrial bioacoustic datasets.	Implements optimized INT8 neural inference on Arduino Nano 33 BLE for edge-side sound recognition.	Uses handcrafted MFCC features which degrade in multipath channels.
7	EdgeVolution: democratizing multi-objective neural architecture search...	Wearable/IoT benchmarks.	Hardware-in-the-loop platform using evolutionary algorithms to deploy neural networks on MCUs.	Requires physical connection during optimization; cannot simulate acoustics.
8	DRL-TinyEdge: Energy- and Latency-Aware DRL for Adaptive TinyML	CIFAR-10 industrial IoT datasets.	Introduces on-device DRL controller to dynamically optimize execution venues and model	Evaluated on terrestrial platforms; lacks specific UASN link simulation.

			config in real time.	
9	TinyML-driven Sensor Nodes for Energy-efficient Acoustic Event Detection in PAWSNs	Environmental acoustic datasets.	Integrates TinyML into pervasive acoustic WSNs, processing audio locally at ~0.45W power limits.	Limited to leaf nodes connected via short-range radio (ESP-Now).
10	A modular embedded architecture for very-low-frequency underwater acoustic acquisition	Experimental hydrophone data.	Integrates a low-noise front-end and Σ - Δ ADC for high-precision VLF (1-1000 Hz) explosion signal capture.	Optimized strictly for transient shock-waves; may underperform on continuous sounds.
11	MO-CRP: An NSGA-II-Based Multiobjective Optimization Cluster Routing Protocol	NS-3 simulated underwater environments.	Develops a multi-objective protocol using NSGA-II to balance energy, battery, load, and coverage.	High computational overhead for MCUs; lacks physical validation.
12	DRL-Based Cooperative Searching of Multi-AUV in Continuous Space...	Simulation environments.	DRL framework (MACSAC) allowing multi-AUVs to cooperatively search continuous space with uncertain regions.	Purely simulation-based; no real ocean drift effects tested.

13	Task Scheduling for Distributed AUV Network Target Hunting... DMAPPO Approach	Simulated turbulent fields.	Proposes DMAPPO scheme based on Age of Information (AoI) for real-time target hunting and environment searching.	High dependence on flawless intra-network acoustic information sharing.
14	Multi-AUV Path Planning Based on Hierarchical Opponent Multi-Actor-Attention-Critic... (HOMAAC)	Bounded/coastal maps.	Integrates geometry-informed task allocation with MARL to achieve up to 98.65% capture success rates.	Assumes simplified obstacle geometries that do not reflect complex reef topography.
15	A multi-agent reinforcement learning algorithm with the action preference selection strategy...	Simulation domains.	Enhances massive target cooperative search mission planning using action preference selection in MARL.	Heavily theoretical; lacks experimental verification on physical AUV fleets.
16	Uncertainty-aware classification of underwater acoustic signals using Bayesian deep learning	Real-world passive sonar and synthetic aperture sonar.	Applies Bayesian Neural Networks to classify signals while estimating aleatoric and epistemic uncertainties.	High computational demand limits edge deployment.

17	Multi-Label Classification of Heterogeneous Underwater Soundscapes with Bayesian Deep Learning	3,400+ hours of passive sonar data.	Uses BDL to accurately classify multiple ships simultaneously present, outputting per-inference uncertainty.	Relies heavily on automatic identification system (AIS) fusion for ground truth.
18	VI-PANN: Harnessing Transfer Learning and Uncertainty-Aware Variational Inference...	ESC-50, UrbanSound8K, DCASE.	Transfers calibrated uncertainty info alongside knowledge from upstream audio tasks to downstream systems.	Evaluated primarily on generalized audio, requiring adaptation for deep-ocean dynamics.
19	Causal inference in environmental sound recognition	Human behavioral experiment datasets.	Shows sound recognition works by inferring physical causes (source power, distance) via reverberation cues.	Cognitive science study; lacks a deployable AI algorithmic framework.
20	iKnow-audio: Integrating Knowledge Graphs with Audio-Language Models	Six benchmark datasets (ESC-50, FSD50K, etc.).	Develops an Audio-centric Knowledge Graph (AKG) to structure semantic, causal, and taxonomic sound relationships.	Focuses on terrestrial sounds; relies on large foundation models.

21	Self-supervised learning-based underwater acoustical signal classification via mask modeling	DeepShip.	Uses mask modeling and a Swin Transformer to reconstruct missing pieces, learning from unlabeled data.	Transformer backbone is too heavy to run on embedded hardware.
22	Time-Frequency Fused Underwater Acoustic Source Localization Based on CPC	Simulated and tank data.	Uses contrastive predictive coding to localize acoustic sources without needing labeled data.	Computationally heavy; hard to run on low-power edge devices.
23	DeepShip: An underwater acoustic benchmark dataset...	DeepShip (benchmark creation).	Introduces 47+ hours of ship engine recordings plus a separable-convolution autoencoder for classification.	Skewed heavily toward coastal ship noise; lacks biological sounds.
24	Underwater Ambient-Noise Removing GAN Based on Magnitude and Phase Spectra	Real/simulated noisy ship data.	Models both magnitude and phase to strip ambient noise (UWAR-GAN), improving SNR by ~45%.	GAN training is unstable; inference is too slow for real-time edge use.

25	Underwater Acoustic Target Classification Based on Dense Convolutional Neural Network	Proprietary vessel data.	Utilizes dense connectivity in CNNs to maximize feature reuse for underwater acoustic target identification.	Subject to standard CNN limitations regarding high parameter count.
26	Modelling of Underwater Vehicles using Physics-Informed Neural Networks with Control	Synthetic trajectory data.	Uses PINNs to bake hydrodynamic constraints into an AI model for robust AUV control.	Assumes simplified hydrodynamics; lacks complex current validation.
27	Physics-Informed Side-Scan Sonar Perception: Tackling Weak Targets... (SADH)	Sonar imaging data (SCTD).	Introduces physics-informed geometric loss to differentiate acoustic shadows of targets from seabed rocks.	Relies strictly on active sonar imagery rather than passive acoustics.
28	Rapid Bayesian Imaging of Large-Scale Geological Surveys	TEM survey data.	Develops a Predictive Neural Network (PNN) outputting posterior PDFs for transient electromagnetics.	Specific to geological imaging rather than acoustic wave detection.

29	Challenges and Advances in Underwater Sonar Systems and AI-Driven Signal Processing...	Review data.	Comprehensive review mapping the evolution of AI-driven signal processing for modern naval sonar systems.	Serves as a review; does not introduce a novel deployable algorithm.
30	Distributed cooperative Kalman filter constrained by advection–diffusion equation...	Mobile sensor network data.	Constrains Kalman filters with advection-diffusion fluid dynamics to track targets in flowing environments.	Performance degrades under highly turbulent, non-laminar flow anomalies.
31	Acoustic Localization of Underwater Robots: A Time of Arrival-Based Particle Filter...	Experimental pool data.	Uses TOA signals with a particle filter to localize robots from asynchronous acoustic beacons.	Needs active beacon pinging; not a passive-listening method.
32	BlueBuzz, an Open-Source Acoustic Modem	Lake testbed data.	Provides an accessible, open-source hardware and software platform for acoustic underwater localization and comms.	Shorter effective range and lower bandwidth compared to commercial modems.

33	CoralBuddy-1: A Micro-Sized Autonomous Underwater Vehicle for Offshore Coral Reef Monitoring	Field testing data in shallow reefs.	Develops a micro-AUV integrating acoustic and visual sensors for offshore coral monitoring.	Restricted to shallow environments with short operational battery life.
34	Deep Learning Prediction of Time-Varying UWA Channel Based on LSTM with Attention	Simulated/real channel data.	LSTM + attention model forecasting acoustic channel changes, aiding pre-adaptation to degrading links.	Tested on specific localized conditions; poor generalizability.
35	Remote Sensing Systems for Ocean: A Review. (Part 2: Active Systems)	Oceanographic sensing datasets.	Details active oceanographic systems including SONAR, altimeters, and High-Frequency radars.	A broad physical review lacking deep edge AI integration specifics.

Summary of Research gaps

The literature identifies several fundamental barriers impeding the widespread operational deployment of intelligent underwater acoustic networks:

- **Absence of Deep-Sea Physical Validation:** Most proposed AI architectures are evaluated exclusively within simulated environments. These simulators fail to adequately replicate the chaotic, real-world variations of ocean sound speed profiles, such as internal waves and sharp thermoclines, rendering their operational reliability unproven.
- **Severe Hardware and Power Constraints at the Edge:** Advanced AI models, including visual Transformers and deep Bayesian networks, impose immense computational burdens. Diverting limited battery capacity to support these complex inferences significantly degrades the mission duration and operational range of subsea micro-nodes.
- **Over-Reliance on Active Acoustic Modalities:** Current methodologies frequently prioritize active sonar perception and beacon pinging for localization and target

detection. This active acoustic approach consumes substantially more energy compared to passive listening and acoustic event detection strategies.

- **Inadequate and Biased Benchmark Datasets:** Existing acoustic datasets are heavily skewed toward coastal anthropogenic noise, particularly commercial shipping. They fail to represent the heterogeneous physical reality of ocean soundscapes, which inherently contain diverse biological and geological acoustic sources.
- **Algorithmic Degradation via Numerical Feature Drift:** Neural networks are typically trained utilizing high-precision 32-bit arithmetic. When these models are quantized to function on low-power 8-bit subsea microcontrollers, structural discrepancies in mathematical libraries induce feature drift, profoundly degrading classification accuracy during live deployments.

Contribution of the study

The following represent a few distinct, advanced technological contributions that could possibly be done:

1. **Event-Triggered Subsea Digital Twin Synchronization Protocol**
Implement an adaptive synchronization mechanism that updates the surface digital twin strictly when a node's residual energy or link quality breaches specific mathematical thresholds. This minimizes acoustic control overhead while preserving real-time network topology mapping.
2. **Thermodynamic Virtual Heat Routing for GPS-Denied Environments**
Formulate a multi-objective routing protocol that constructs a virtual "heat field" based on node depth, hop count, and residual energy. This framework allows data packets to naturally propagate toward high-capacity surface sinks, eliminating the need for complex, energy-intensive 3D spatial localization coordinates.
3. **Physics-Informed Neural Network (PINN) Controllers for Autonomous Navigation**
Embed hydrodynamic differential equations directly into the loss function of the neural controller. This mathematically restricts the AI from predicting physically impossible state transitions, drastically improving vehicle stability when encountering undocumented ocean turbulence.
4. **Phase-Aware Adversarial Denoising Architecture for Acoustic Modems**
Engineer a Generative Adversarial Network (GAN) tailored for the physical acoustic layer. By processing both log magnitudes and complex phase spectra, the model can mathematically reconstruct degraded signals, reversing severe multipath distortion and removing ambient noise in real time.

5. **Uncertainty-Aware Bayesian Edge Classifiers for Autonomous Decision-Making**
Develop a Bayesian Deep Learning payload capable of outputting calibrated uncertainty scores alongside its multi-label classifications. By quantifying its own aleatoric and epistemic uncertainty, the autonomous vehicle can dynamically opt to acquire higher-quality acoustic samples before executing a critical command.
6. **Continuous-Space Multi-Agent Reinforcement Learning (MARL) for Swarm Coordination**
Design a MARL framework utilizing probabilistic Gaussian distributions to navigate continuous space, discarding energy-inefficient grid-based discretizations. This permits multi-vehicle swarms to optimize path planning and dynamically alternate between target hunting and environmental mapping.
7. **Hardware-in-the-Loop Neural Architecture Search for Subsea Microcontrollers**
Automate the multi-objective optimization of subsea algorithms by deploying candidate models directly onto target Microcontroller Units (MCUs) during the evolutionary search phase. This ensures the resulting models strictly adhere to the operational latency and static RAM limitations of the physical hardware.
8. **Zero-Drift Quantization Pipelines for Edge Acoustics**
Construct a rigorous cross-platform compilation methodology that enforces perfect alignment of acoustic feature extraction parameters between training environments and edge deployments. This pipeline will neutralize numerical drift during 8-bit quantization, preserving the precision of TinyML hydroacoustic classifiers.

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